



Python

```
import matplotlib
matplotlib.use("Agg") # headless: every chart is
    saved to file, not popped up
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
import os

sns.set_style("whitegrid")

OUT_DIR = "dashboard_charts"
os.makedirs(OUT_DIR, exist_ok=True)

#
-----
# Load data (adjust the path if the CSV is elsewhere)
#
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df = pd.read_csv("Employee_Data_Analyst.csv",
    skiprows=2)
df = df.drop(columns=[c for c in df.columns if
    'Unnamed' in c])
df.columns = [c.strip() for c in df.columns]

role_order = ['Intern', 'Analyst', 'Team Lead',
    'Manager', 'Executive']

def save(fig, name):
    fig.tight_layout()
    fig.savefig(os.path.join(OUT_DIR, name), dpi=150,
        bbox_inches='tight')
    plt.close(fig)

# 1. Bar - employee count by Department
fig, ax = plt.subplots(figsize=(6, 4.5))
dept_counts = df['Department'].value_counts()
ax.bar(dept_counts.index, dept_counts.values,
    color='steelblue')
ax.set_title("Employees by Department")
ax.tick_params(axis='x', rotation=45)
save(fig, "01_bar_department.png")

# 2. Bar - employee count by JobRole
fig, ax = plt.subplots(figsize=(6, 4.5))
role_counts =
    df['JobRole'].value_counts().reindex(role_order)
ax.bar(role_counts.index, role_counts.values,
    color='darkorange')
ax.set_title("Employees by Job Role")
ax.tick_params(axis='x', rotation=45)
save(fig, "02_bar_jobrole.png")

# 3. Grouped bar - Attrition by Department
fig, ax = plt.subplots(figsize=(7, 4.5))
grp = df.groupby(['Department',
    'Attrition']).size().unstack(fill_value=0)
x = np.arange(len(grp.index))
width = 0.35
ax.bar(x - width/2, grp['Yes'], width, label='Yes',
    color='indianred')
ax.bar(x + width/2, grp['No'], width, label='No',
    color='seagreen')
ax.set_xticks(x); ax.set_xticklabels(grp.index,
    rotation=45)
ax.set_title("Attrition by Department (Grouped)")
ax.legend(title="Attrition")
save(fig, "03_grouped_bar_attrition.png")

# 4. Stacked bar - Gender by Department
fig, ax = plt.subplots(figsize=(7, 4.5))
gstack = df.groupby(['Department',
    'Gender']).size().unstack(fill_value=0)
bottom = np.zeros(len(gstack))
for gender in gstack.columns:
    ax.bar(gstack.index, gstack[gender], bottom=bottom,
        label=gender)
    bottom += gstack[gender].values
ax.set_title("Gender Mix by Department (Stacked)")
ax.tick_params(axis='x', rotation=45)
ax.legend(title="Gender")
save(fig, "04_stacked_bar_gender.png")

# 5. Pie - Gender distribution
fig, ax = plt.subplots(figsize=(6, 6))
gender_counts = df['Gender'].value_counts()
ax.pie(gender_counts.values,
    labels=gender_counts.index, autopct='%1.0f%%',
    colors=sns.color_palette('Set2'))
ax.set_title("Gender Distribution")
save(fig, "05_pie_gender.png")

# 6. Pie - Education Level distribution
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R

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library(ggplot2)
library(dplyr)
library(tidyrr)
library(reshape2)

dir.create("dashboard_charts", showWarnings = FALSE)

#
-----
# Load data
#
-----
df <- read.csv("Employee_Data_Analyst.csv", skip = 2)
df <- df[, !grepl("^X", names(df))]
names(df) <- trimws(names(df))
names(df) <- gsub("\\.", "", names(df))
names(df) <- trimws(names(df))
names(df)[names(df) == "Education Level"] <-
    "EducationLevel"
names(df)[names(df) == "Years At Company"] <-
    "YearsAtCompany"
names(df)[names(df) == "Monthly Income"] <-
    "MonthlyIncome"
names(df)[names(df) == "Job Satisfaction"] <-
    "JobSatisfaction"
names(df)[names(df) == "Work Life Balance"] <-
    "WorkLifeBalance"
names(df)[names(df) == "Performance Rating"] <-
    "PerformanceRating"
names(df)[names(df) == "Distance From Home"] <-
    "DistanceFromHome"
names(df)[names(df) == "Training Times Last Year"] <-
    "TrainingTimesLastYear"
names(df)[names(df) == "Num Projects"] <-
    "NumProjects"

role_order <- c('Intern', 'Analyst', 'Team Lead',
    'Manager', 'Executive')

save_plot <- function(p, name, width = 6, height =
    4.5) {
  ggsave(file.path("dashboard_charts", name), plot = p,
    width = width, height = height, dpi = 150)
}

# 1. Bar - employees by Department
dept_counts <- as.data.frame(table(df$Department))
colnames(dept_counts) <- c('Department', 'Count')
p1 <- ggplot(dept_counts, aes(x=Department, y=Count))
+
  geom_col(fill="steelblue") + labs(title="Employees by
    Department") +
  theme_minimal(base_size=13) +
    theme(axis.text.x=element_text(angle=45, hjust=1))
save_plot(p1, "01_bar_department.png")

# 2. Bar - employees by Job Role
role_counts <- as.data.frame(table(factor(df$JobRole,
    levels=role_order)))
colnames(role_counts) <- c('JobRole', 'Count')
p2 <- ggplot(role_counts, aes(x=JobRole, y=Count)) +
  geom_col(fill="darkorange") + labs(title="Employees by
    Job Role") +
  theme_minimal(base_size=13) +
    theme(axis.text.x=element_text(angle=45, hjust=1))
save_plot(p2, "02_bar_jobrole.png")

# 3. Grouped bar - Attrition by Department
grp <- df %>% count(Department, Attrition)
p3 <- ggplot(grp, aes(x=Department, y=n,
    fill=Attrition)) +
  geom_col(position=position_dodge()) +
  scale_fill_manual(values=c(Yes="indianred",
    No="seagreen")) +
  labs(title="Attrition by Department (Grouped)",
    y="Count") +
  theme_minimal(base_size=13) +
    theme(axis.text.x=element_text(angle=45, hjust=1))
save_plot(p3, "03_grouped_bar_attrition.png", width=7)

# 4. Stacked bar - Gender by Department
gstack <- df %>% count(Department, Gender)
p4 <- ggplot(gstack, aes(x=Department, y=n,
    fill=Gender)) +
  geom_col() + labs(title="Gender Mix by Department
    (Stacked)", y="Count") +
  theme_minimal(base_size=13) +
    theme(axis.text.x=element_text(angle=45, hjust=1))
save_plot(p4, "04_stacked_bar_gender.png", width=7)

# 5. Pie - Gender distribution
gender_counts <- as.data.frame(table(df$Gender))
colnames(gender_counts) <- c('Gender', 'Count')
p5 <- ggplot(gender_counts, aes(x="", y=Count,
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fig, ax = plt.subplots(figsize=(6, 6))
edu_counts = df['Education Level'].value_counts()
ax.pie(edu_counts.values, labels=edu_counts.index,
       autopct='%1.0P%%',
       colors=sns.color_palette('Set3'))
ax.set_title("Education Level Distribution")
save(fig, "06_pie_education.png")

# 7. Histogram - Age distribution
fig, ax = plt.subplots(figsize=(6, 4.5))
ax.hist(df['Age'], bins=15, color='mediumpurple',
       edgecolor='black')
ax.set_title("Age Distribution")
ax.set_xlabel("Age")
save(fig, "07_hist_age.png")

# 8. Histogram - Monthly Income distribution
fig, ax = plt.subplots(figsize=(6, 4.5))
ax.hist(df['Monthly Income'], bins=20,
       color='darkcyan', edgecolor='black')
ax.set_title("Monthly Income Distribution")
ax.set_xlabel("Monthly Income")
save(fig, "08_hist_income.png")

# 9. Box plot - Monthly Income by Department
fig, ax = plt.subplots(figsize=(7, 4.5))
sns.boxplot(data=df, x='Department', y='Monthly Income', hue='Department',
            palette='Set2', legend=False, ax=ax)
ax.set_title("Income Spread by Department")
ax.tick_params(axis='x', rotation=45)
save(fig, "09_box_income_department.png")

# 10. Violin plot - Job Satisfaction by Attrition
fig, ax = plt.subplots(figsize=(6, 4.5))
sns.violinplot(data=df, x='Attrition',
              y='JobSatisfaction', hue='Attrition',
              palette='Set3', legend=False, ax=ax)
ax.set_title("Job Satisfaction by Attrition")
save(fig, "10_violin_satisfaction.png")

# 11. Scatter - Age vs Monthly Income
fig, ax = plt.subplots(figsize=(6, 4.5))
ax.scatter(df['Age'], df['Monthly Income'], alpha=0.4,
          color='darkorange', s=20)
ax.set_title("Age vs Monthly Income")
ax.set_xlabel("Age"); ax.set_ylabel("Income")
save(fig, "11_scatter_age_income.png")

# 12. Bubble chart - Years At Company vs Income,
size=NumProjects, color=Attrition
fig, ax = plt.subplots(figsize=(7, 4.5))
colors_map = {'Yes': 'firebrick', 'No': 'steelblue'}
for att, sub in df.groupby('Attrition'):
    ax.scatter(sub['Years At Company'], sub['Monthly Income'],
              s=sub['Num Projects'] * 10, alpha=0.4,
              color=colors_map[att], label=att)
ax.set_title("Tenure vs Income (bubble = Num Projects)")
ax.set_xlabel("Years At Company");
ax.set_ylabel("Income")
ax.legend(title="Attrition")
save(fig, "12_bubble_tenure_income.png")

# 13. Correlation heatmap - numeric variables
fig, ax = plt.subplots(figsize=(7.5, 6.5))
numeric_cols = ['Age', 'Years At Company', 'Monthly Income', 'JobSatisfaction', 'Work Life Balance', 'Performance Rating', 'Distance From Home', 'Training Times Last Year', 'Num Projects']
corr = df[numeric_cols].corr()
sns.heatmap(corr, cmap='coolwarm', center=0, ax=ax,
            annot=True, fmt='.2f',
            annot_kws={'size': 7}, cbar_kws={'shrink': 0.8})
ax.set_title("Correlation Heatmap")
ax.tick_params(axis='x', rotation=45, labels=8)
ax.tick_params(axis='y', labels=8)
save(fig, "13_heatmap_correlation.png")

# 14. Density plot - Monthly Income by Gender
fig, ax = plt.subplots(figsize=(6, 4.5))
for gender in df['Gender'].unique():
    sns.kdeplot(df[df['Gender'] == gender]['Monthly Income'], ax=ax, label=gender, fill=True,
               alpha=0.25)
ax.set_title("Income Density by Gender")
ax.legend(title="Gender")
save(fig, "14_density_income_gender.png")

# 15. Line chart - Average Income by Years At Company
fig, ax = plt.subplots(figsize=(6, 4.5))
avg_income = df.groupby('Years At Company')['Monthly Income'].mean().sort_index()
ax.plot(avg_income.index, avg_income.values,
       marker='o', color='seagreen')
ax.set_title("Avg Income by Tenure")
ax.set_xlabel("Years At Company");
ax.set_ylabel("Average Income")

fill=Gender)) +
geom_col(width=1) + coord_polar("y") +
labs(title="Gender Distribution") +
theme_void(base_size=13)
save_plot(p5, "05_pie_gender.png", width=6, height=6)

# 6. Pie - Education Level distribution
edu_counts <- as.data.frame(table(df$EducationLevel))
colnames(edu_counts) <- c('EducationLevel', 'Count')
p6 <- ggplot(edu_counts, aes(x="", y=Count,
fill=EducationLevel)) +
geom_col(width=1) + coord_polar("y") +
labs(title="Education Level Distribution") +
theme_void(base_size=13)
save_plot(p6, "06_pie_education.png", width=6,
height=6)

# 7. Histogram - Age
p7 <- ggplot(df, aes(x=Age)) +
geom_histogram(bins=15, fill="mediumpurple",
color="black") +
labs(title="Age Distribution") +
theme_minimal(base_size=13)
save_plot(p7, "07_hist_age.png")

# 8. Histogram - Monthly Income
p8 <- ggplot(df, aes(x=MonthlyIncome)) +
geom_histogram(bins=20, fill="darkcyan",
color="black") +
labs(title="Monthly Income Distribution") +
theme_minimal(base_size=13)
save_plot(p8, "08_hist_income.png")

# 9. Box plot - Income by Department
p9 <- ggplot(df, aes(x=Department, y=MonthlyIncome,
fill=Department)) +
geom_boxplot(show.legend=FALSE) + labs(title="Income Spread by Department") +
theme_minimal(base_size=13) +
theme(axis.text.x=element_text(angle=45, hjust=1))
save_plot(p9, "09_box_income_department.png", width=7)

# 10. Violin - Job Satisfaction by Attrition
p10 <- ggplot(df, aes(x=Attrition, y=JobSatisfaction,
fill=Attrition)) +
geom_violin(show.legend=FALSE) + labs(title="Job Satisfaction by Attrition") +
theme_minimal(base_size=13)
save_plot(p10, "10_violin_satisfaction.png")

# 11. Scatter - Age vs Monthly Income
p11 <- ggplot(df, aes(x=Age, y=MonthlyIncome)) +
geom_point(alpha=0.4, color="darkorange") +
labs(title="Age vs Monthly Income") +
theme_minimal(base_size=13)
save_plot(p11, "11_scatter_age_income.png")

# 12. Bubble - Tenure vs Income, size=NumProjects,
color=Attrition
p12 <- ggplot(df, aes(x=YearsAtCompany,
y=MonthlyIncome, size=NumProjects, color=Attrition)) +
geom_point(alpha=0.4) +
scale_color_manual(values=c(Yes="firebrick",
No="steelblue")) +
labs(title="Tenure vs Income (bubble = Num Projects)") +
theme_minimal(base_size=13)
save_plot(p12, "12_bubble_tenure_income.png", width=7)

# 13. Correlation heatmap - numeric variables
numeric_cols <-
c('Age','YearsAtCompany','MonthlyIncome','JobSatisfac
tion',
'WorkLifeBalance','PerformanceRat
ing','DistanceFromHome',
'TrainingTimesLastYear','NumProjects')
corr_mat <- round(corr(df[, numeric_cols]), 2)
corr_long <- melt(corr_mat)
p13 <- ggplot(corr_long, aes(x=Var1, y=Var2,
fill=value)) +
geom_tile() + geom_text(aes(label=value), size=3) +
scale_fill_gradient2(low="blue", mid="white",
high="red", midpoint=0) +
labs(title="Correlation Heatmap", x="", y="") +
theme_minimal(base_size=12) +
theme(axis.text.x=element_text(angle=45, hjust=1))
save_plot(p13, "13_heatmap_correlation.png",
width=7.5, height=6.5)

# 14. Density - Income by Gender
p14 <- ggplot(df, aes(x=MonthlyIncome, fill=Gender)) +
geom_density(alpha=0.3) + labs(title="Income Density by Gender") +
theme_minimal(base_size=13)
save_plot(p14, "14_density_income_gender.png")

# 15. Line - Avg Income by Tenure
avg_income <- df %>% group_by(YearsAtCompany) %>%

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<pre> save(fig, "15_line_avg_income_tenure.png") # 16. Stacked area - Attrition proportion across tenure bins fig, ax = plt.subplots(figsize=(6, 4.5)) df['TenureBin'] = pd.cut(df['Years At Company'], bins=[-1, 3, 6, 9, 12, 15], labels=['0-3', '4-6', '7-9', '10-12', '13-15']) prop = df.groupby(['TenureBin', 'Attrition'], observed=True).size().unstack(fill_value=0) prop = prop.div(prop.sum(axis=1), axis=0) ax.stackplot(prop.index.astype(str), prop['Yes'], prop['No'], labels=['Yes', 'No'], colors=['indianred', 'seagreen']) ax.set_title("Attrition Proportion by Tenure Bin") ax.legend(title="Attrition") ax.set_ylim(0, 1) save(fig, "16_stacked_area_attrition.png") # 17. Treemap - Department headcount share (manual, no extra dependency) fig, ax = plt.subplots(figsize=(7, 3.5)) dept_totals = df['Department'].value_counts() sizes = dept_totals.values labels = dept_totals.index norm_sizes = sizes / sizes.sum() x_cursor = 0 colors_tm = sns.color_palette('Set2', len(labels)) for lbl, sz, col in zip(labels, norm_sizes, colors_tm): ax.barh(0, sz, left=x_cursor, color=col, edgecolor='white') ax.text(x_cursor + sz / 2, 0, f'{lbl}\n{int(sz*sizes.sum())}', ha='center', va='center', fontsize=9) x_cursor += sz ax.set_xlim(0, 1); ax.set_ylim(-1, 1) ax.axis('off') ax.set_title("Department Headcount Share (Treemap)") save(fig, "17_treemap_department.png") # 18. Radar chart - Avg JobSatisfaction / WorkLifeBalance / PerformanceRating by Department fig = plt.figure(figsize=(6.5, 6.5)) ax = fig.add_subplot(111, projection='polar') metrics = ['JobSatisfaction', 'Work Life Balance', 'Performance Rating'] angles = np.linspace(0, 2 * np.pi, len(metrics), endpoint=False).tolist() angles += angles[:1] for dept in df['Department'].unique()[1:4]: # limit to 4 depts to keep it legible vals = df[df['Department'] == dept][metrics].mean().tolist() vals += vals[:1] ax.plot(angles, vals, linewidth=1.5, label=dept) ax.fill(angles, vals, alpha=0.08) ax.set_xticks(angles[:1]) ax.set_xticklabels(metrics) ax.set_title("Dept Avg Ratings (Radar)", pad=25) ax.legend(loc='upper right', bbox_to_anchor=(1.35, 1.1)) save(fig, "18_radar_department_ratings.png") # 19. Funnel - Job Role hierarchy headcount fig, ax = plt.subplots(figsize=(6.5, 4.5)) funnel_counts = df['JobRole'].value_counts().reindex(role_order) max_val = funnel_counts.max() for i, (role, cnt) in enumerate(funnel_counts.items()): ax.barh(i, cnt, left=-cnt / 2, height=0.6, color=sns.color_palette('Blues', len(role_order))[i]) ax.text(0, i, f'{role}; {cnt}', ha='center', va='center', fontsize=10) ax.set_yticks([]) ax.set_xlim(-max_val / 2 - 10, max_val / 2 + 10) ax.set_title("Job Role Funnel") save(fig, "19_funnel_jobrole.png") # 20. ECDF - Monthly Income fig, ax = plt.subplots(figsize=(6, 4.5)) sorted_income = np.sort(df['Monthly Income']) ecdf_y = np.arange(1, len(sorted_income) + 1) / len(sorted_income) ax.plot(sorted_income, ecdf_y, color='darkorange') ax.set_title("ECDF - Monthly Income") ax.set_xlabel("Monthly Income") ax.set_ylabel("Cumulative Proportion") save(fig, "20_ecdf_income.png") print(f'All 20 charts saved individually to ./{OUT_DIR}/ - no overlap, no shared cramped grid.") </pre>	<pre> summarise(MonthlyIncome=mean(MonthlyIncome)) p15 <- ggplot(avg_income, aes(x=YearsAtCompany, y=MonthlyIncome)) + geom_line(color='seagreen') + geom_point(color='seagreen') + labs(title="Avg Income by Tenure") + theme_minimal(base_size=13) save_plot(p15, "15_line_avg_income_tenure.png") # 16. Stacked area - Attrition proportion by tenure bin df\$TenureBin <- cut(df\$YearsAtCompany, breaks=c(-1,3,6,9,12,15), labels=c('0-3','4-6','7-9','10-12','13-15')) prop <- df %>% count(TenureBin, Attrition) %>% group_by(TenureBin) %>% mutate(Proportion = n / sum(n)) p16 <- ggplot(prop, aes(x=TenureBin, y=Proportion, fill=Attrition, group=Attrition)) + geom_area(position='stack') + scale_fill_manual(values=c(Yes="indianred", No="seagreen")) + labs(title="Attrition Proportion by Tenure Bin") + theme_minimal(base_size=13) save_plot(p16, "16_stacked_area_attrition.png") # 17. Treemap - Department headcount share (manual geom_rect) dept_tot <- df %>% count(Department) %>% arrange(desc(n)) %>% mutate(frac = n / sum(n), xmax = cumsum(frac), xmin = lag(xmax, default=0)) p17 <- ggplot(dept_tot) + geom_rect(aes(xmin=xmin, xmax=xmax, ymin=0, ymax=1, fill=Department), color="white") + geom_text(aes(x=(xmin+xmax)/2, y=0.5, label=paste0(Department, "n", n)), size=3.5) + labs(title="Department Headcount Share (Treemap)") + theme_void(base_size=13) + theme(legend.position="none") save_plot(p17, "17_treemap_department.png", width=7, height=3.5) # 18. Radar - Avg ratings by Department (first 4 departments) metrics <- c('JobSatisfaction', 'WorkLifeBalance', 'PerformanceRating') top_depts <- unique(df\$Department)[1:4] radar_df <- df %>% filter(Department %in% top_depts) %>% group_by(Department) %>% summarise(across(all_of(metrics), mean)) %>% pivot_longer(cols=all_of(metrics), names_to="Metric", values_to="Value") p18 <- ggplot(radar_df, aes(x=Metric, y=Value, group=Department, color=Department)) + geom_polygon(fill=NA, linewidth=0.8) + geom_point(size=2) + coord_polar() + labs(title="Dept Avg Ratings (Radar)", x="", y="") + theme_minimal(base_size=12) save_plot(p18, "18_radar_department_ratings.png", width=6.5, height=6.5) # 19. Funnel - Job Role hierarchy headcount (manual, centered bars) funnel_df <- as.data.frame(table(factor(df\$JobRole, levels=role_order))) colnames(funnel_df) <- c('JobRole', 'Count') p19 <- ggplot(funnel_df, aes(x=JobRole, y=Count)) + geom_col(aes(y=Count), fill="steelblue", width=0.6) + geom_col(aes(y=Count), fill="steelblue", width=0.6) + geom_text(aes(y=0, label=paste0(JobRole, ": ", Count)), size=3.5) + coord_flip() + labs(title="Job Role Funnel", x="", y="") + theme_minimal(base_size=13) + theme(axis.text.y=element_blank()) save_plot(p19, "19_funnel_jobrole.png", width=6.5) # 20. ECDF - Monthly Income p20 <- ggplot(df, aes(x=MonthlyIncome)) + stat_ecdf(color="darkorange") + labs(title="ECDF - Monthly Income", y="Cumulative Proportion") + theme_minimal(base_size=13) save_plot(p20, "20_ecdf_income.png") cat("All 20 charts saved individually to ./dashboard_charts/ - no overlap, no shared cramped grid.\n") </pre>
Output	Output













